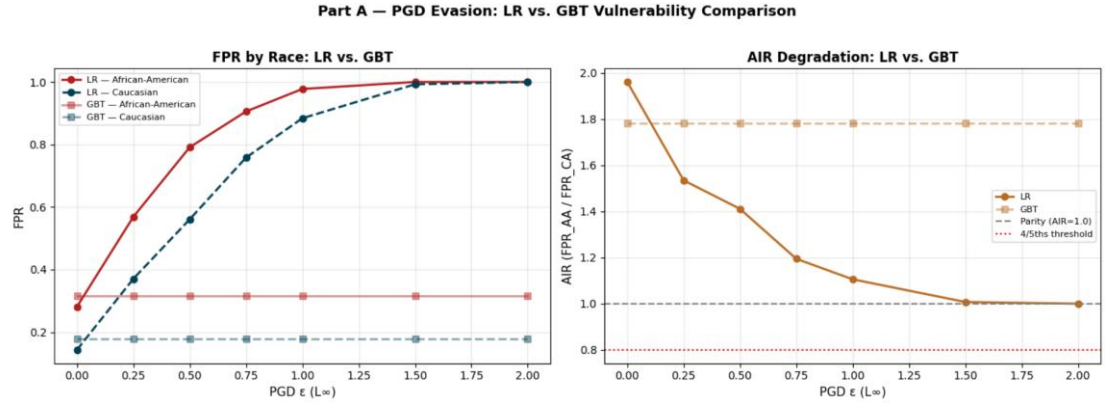
Individual Assignment 5 G49654949 LinlinChen

Part 1: PGD Evasion Audit

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AIR crossover summary (ε at which AIR < 0.80):
  LR: AIR does not fall below 0.80 in tested range.
  GBT: AIR does not fall below 0.80 in tested range.
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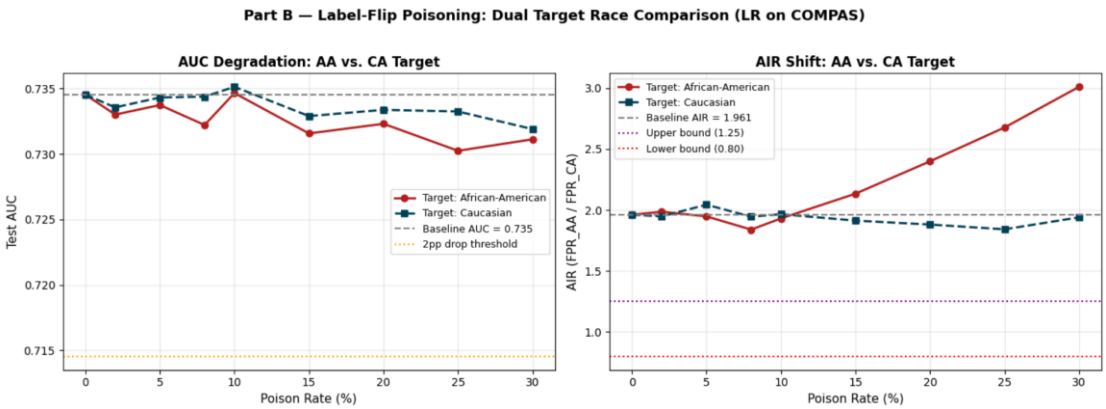
The two models exhibit fundamentally different vulnerability profiles under the PGD evasion attack. LR model is highly vulnerable to PGD evasion: when ε = 0.25, African-American FPR jumps from 0.281 to 0.569, AIR drops sharply from 1.961 to 1.535, and by ε = 2.0 the FPR for both groups saturates at 1.000 with an AIR of 1.000. In contrast, the GBT model is effectively immune to the gradient-based PGD attack, both FPR and AIR remain unchanged across all ε values, because finite-difference gradient estimates are too noisy on this 7-feature tabular dataset to generate effective perturbations. For high-stakes deployments, this implies the GBT is the safer choice against evasion attacks, while LR requires input validation or adversarial training before deployment.

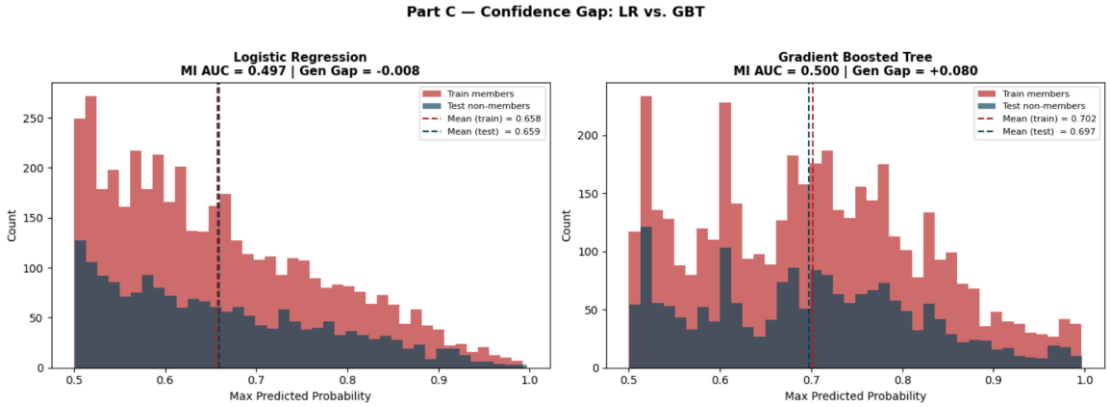
Part 2: Poisoning Loop with Fairness Monitoring

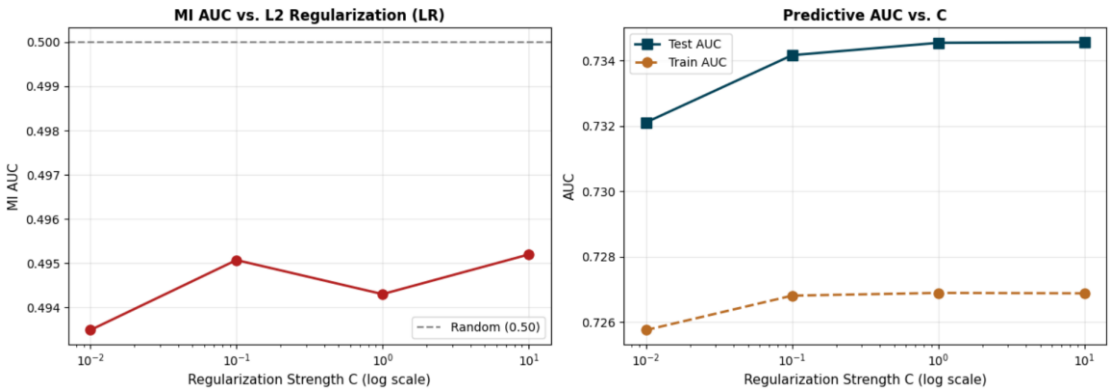


The above figure shows AUC and AIR degradation curves for both target-race variants (African-American and Caucasian) as the poison rate increases from 0% to 30%. For AUC panel, both curves stay flat within 2 pp of the baseline (0.735) across all poison rates, indicating that the attack is invisible to accuracy-based monitoring. The stealth zone spans 0%–30% for both variants because the baseline model already has AIR = 1.961, which exceeds the upper bound of 1.25 from the start, the fairness threshold is violated before any attack begins. This means the PSI monitor would be perpetually alarmed even on clean data, making the [0.80, 1.25] criterion meaningless here. The PSI-based monitor fails to detect

either attack at any poison rate. Label-flip poisoning only alters a small subgroup's labels, so the aggregate PSI shift remains far below 0.10. Effective detection would require per-subgroup fairness monitoring rather than aggregate PSI.

Part 3: Membership Inference Depth





Both models show members and non-members with heavily overlapping confidence distributions, meaning neither model leaks meaningful membership information. The theoretical prediction holds well for LR (predicted 0.496, actual 0.497), but overestimates for GBT (predicted 0.540, actual 0.500). The discrepancy occurs because the meta-classifier was trained on LR shadow. Stronger L2 regularization with lower C slightly reduces MI AUC and test AUC. For meaningful privacy protection, differential privacy or data minimization would be more effective than L2 regularization alone.

Part 4: Reflection

The Highest-risk finding is label-flip poisoning attack is the most critical risk, it degrades fairness silently, evades PSI monitoring, and exploits the pre-existing racial disparity as cover. As a proactive mitigation, implementing per-subgroup FPR monitoring with a ±5 pp alert threshold would flag AA-targeted poisoning as early as an 8% poison rate (FPR_AA drops 6.8 pp), whereas aggregate PSI never triggers, though symmetric thresholds must be applied to all groups equally to avoid differential scrutiny. As a reactive mitigation, retraining with stronger L2 regularization (C = 0.01) yields AIR = 1.966, nearly identical to the baseline (1.961), confirming no additional fairness harm while marginally reducing MI AUC.